

SHETH L.U.J AND SIR M.V COLLEGE

The screenshot shows the RStudio interface with the following components:

- Source:** Contains the R script code for loading data, performing a two-way ANOVA, and summarizing the results.
- Console:** Displays the output of the R script, including the ANOVA summary table and the printed file path.
- Environment:** Lists the objects in the global environment, including the loaded data files and the ANOVA model.
- Files:** Shows the file explorer with the project directory and the data files.

```
> # --- 1. Load Libraries ---
> library(stats)
>
> # --- 2. Load and Prepare Data ---
> crimes_df <- read.csv("C:/Users/yukta/Downloads/crimes_against_women_2001-2014.csv")
>
> # Convert State and Year to factors (categories) so R treats them as groups
> crimes_df$STATE.UT <- as.factor(crimes_df$STATE.UT)
> crimes_df$Year <- as.factor(crimes_df$Year)
>
> # --- 3. Perform Two-way ANOVA ---
> # Formula: Dependent_Variable ~ Factor1 + Factor2
> two_way_model <- aov(Rape ~ STATE.UT + Year, data = crimes_df)
>
> # --- 4. Result & Explanation ---
> summary(two_way_model)
>
  Sum Sq Mean Sq F value Pr(>F)
STATE.UT    71  15124544   213022  4.755 <2e-16 ***
Year        12    320124    26677   0.596  0.848
Residuals 10593  47452672    44796
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> print("Yukta Sonawane S120")
[1] "Yukta Sonawane S120"
> |
```

The ANOVA summary table shows the following results:

	Sum Sq	Mean Sq	F value	Pr(>F)
STATE.UT	71	15124544	213022	4.755 <2e-16 ***
Year	12	320124	26677	0.596 0.848
Residuals	10593	47452672	44796	

The console output also shows the printed file path: [1] "Yukta Sonawane S120".